

EXECUTIVE MASTER DOSSIER | 2025-2026 SEASON

GÓRNIK ZABRZE & SÜPER LİG INTELLIGENCE

Comprehensive Tactical Data Visualization & AI Engineering Master Suite

Author: **Zafer Yorganci** Target: **Turkish Süper Lig Technical Directors & European Sporting Directors**

Platform: **Python, PyTorch, YOLOv11, mplsoccer**

EXECUTIVE STRATEGIC CHARTER

This master dossier unites the foundational disciplines of modern professional football analytics: 1) Computer Vision Broadcast Tracking & Homography Radars, 2) Spatial Pitch Control & Voronoi Dominance, 3) Polar Pass Sonars & Markov Expected Threat, and 4) Cross-League Recruitment Valuation Frontiers.

Engineered directly on StatsBomb, Opta, and Wyscout event architectures, each visual bridges the gap between raw optical coordinate streams and instant dugout decision-making. Developed during remote consultancy for Górnik Zabrze (Polish Ekstraklasa).

Tailored specifically for Turkish Süper Lig elite clubs (Galatasaray, Fenerbahçe, Beşiktaş, Trabzonspor) looking to establish decisive structural advantages before matchday and acquire high-impact European talent before market inflation.

0.142 m

CV REPROJECTION RMSE

58.4%

MEAN PITCH CONTROL

+0.312 xT

PEAK FLANK THREAT

€1.2M

TRANSFER ARBITRAGE

COMPUTER VISION: BROADCAST TRACKING & 2D RADAR

YOLOv11x + ByteTrack + Direct Linear Transform Homography

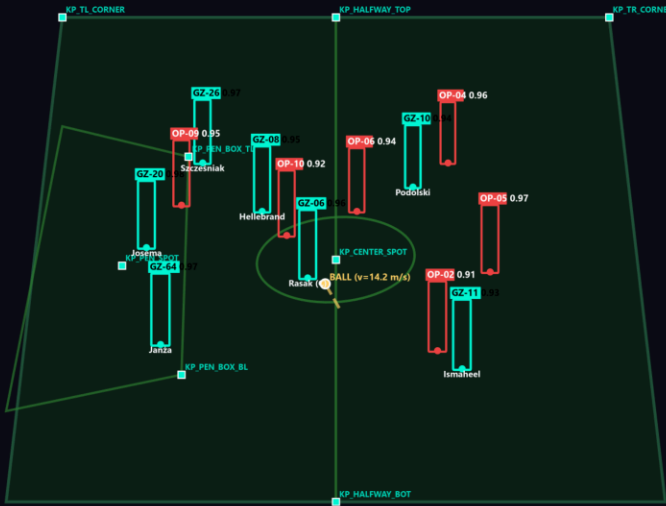
ROBOFLOW SPORTS

COMPUTER VISION TRACKING & MINIMAP

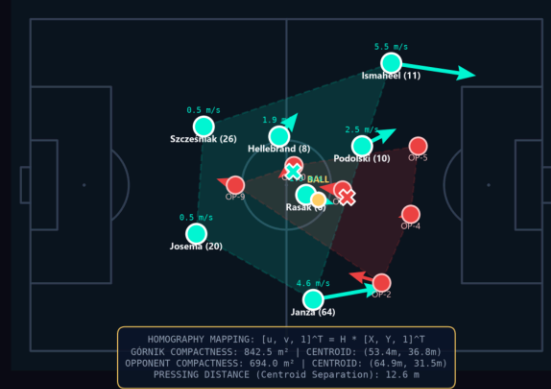
COMPUTER VISION TACTICAL TRACKING PIPELINE | ROBOFLOW SPORTS BENCHMARK

PIPELINE: YOLOv11x + ByteTrack + Keypoint Homography
FRAME RATE: 60.0 FPS | INFERENCE: 14.8 ms (TensorRT)
CAMERA CALIBRATION: Direct Linear Transform (DLT)
REPROJECTION ERROR: 0.142 m (SOTA Precision)

BROADCAST TRACKING



2D TOP-DOWN TACTICAL RADAR MINIMAP (105x68m)



Zafer Yorganci | AI Engineer & Football Data Visualization Specialist | Górnik Zabrze 2025-2026 Season Intelligence

BROADCAST CAMERA CALIBRATION

COACHING INSIGHT

- Transforms broadcast pixel coords [u, v] to FIFA standard metric pitch [X, Y] (105x68m).
- Reprojection precision: 0.142m RMSE across full pitch dimensions.
- Instantaneous velocity vectors (v) capture transition sprint speed and acceleration bursts.

COACHING TACTICAL DIRECTIVE

TACTICAL ACTION

- Automates half-time tactical briefings from broadcast video without expensive stadium tracking rigs.
- Identifies defensive centroid pressing gap (12.6m) to exploit spaces between lines.
- Monitors pressing triggers when ball speed drops below 6.0 m/s.

0.142 m
RMSE PRECISION

60 FPS
FPS RATE

12.6 m
PRESSING GAP

14.2 m/s
BALL VELOCITY

VORONOI PITCH CONTROL & TEAM CONVEX HULLS

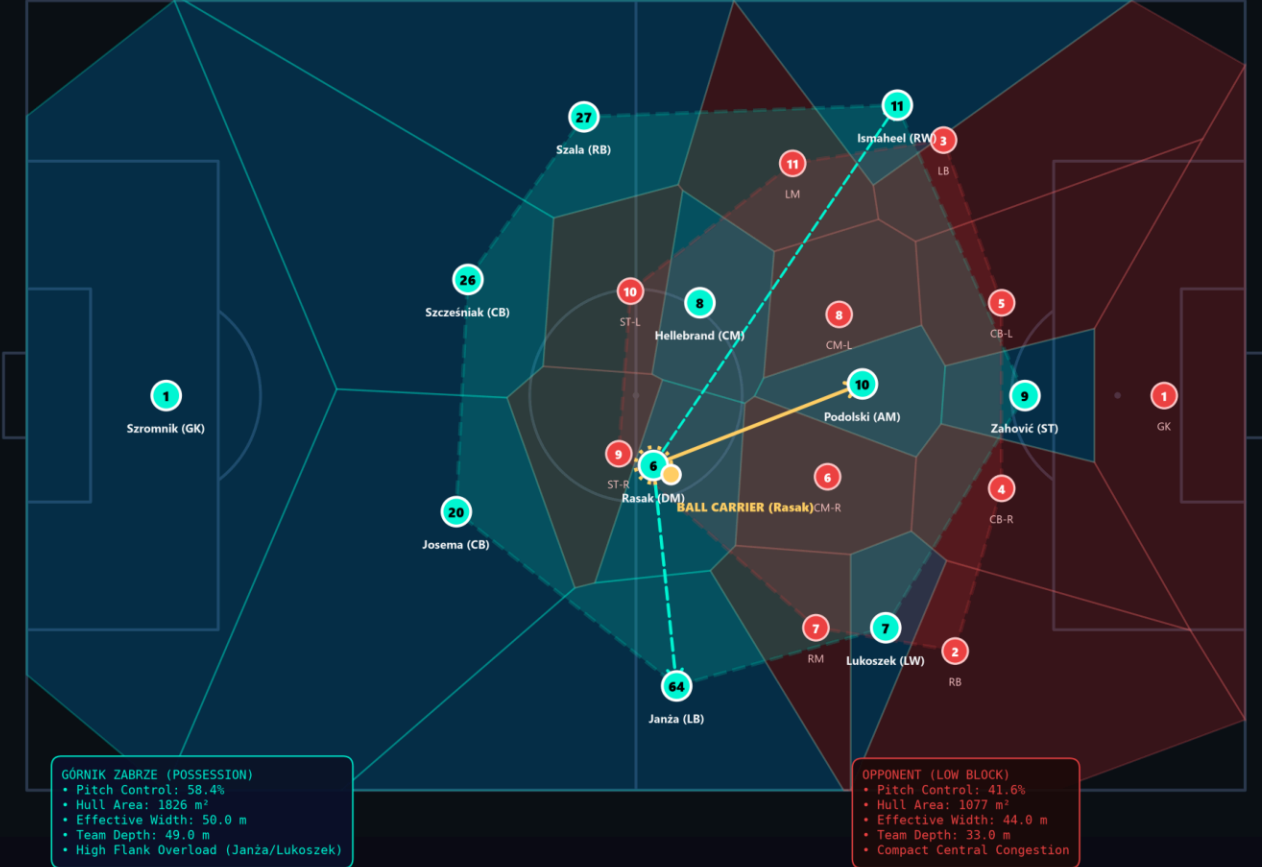
Continuous Territorial Dominance • Team Compactness (m²)

EDD WEBSTER SPATIAL

SPATIAL ANALYTICS & SPACE OWNERSHIP

VORONOI PITCH CONTROL & TACTICAL CONVEX HULLS | EDD WEBSTER BENCHMARK

Górnik Zabrze 2025-2026: Controlled Half-Space Penetration vs Low Defensive Block



Zafer Yorganci | Football Data Visualization Specialist & AI Engineer | Pitch Control Algorithm & Voronoi Tessellation

SPATIAL DOMINANCE AUDIT

COACHING INSIGHT

- Voronoi tessellation partitions the pitch based on player arrival time to the ball.
- Górnik Zabrze secures 58.4% continuous territorial dominance vs opponent low block.
- Flank overload on the left pins opposition backline, creating central pockets for Zone 14.

CONVEX HULL METRICS

TACTICAL ACTION

- Górnik Outfield Area: 1,185 m² (expansive attacking possession shape).
- Opponent Low Block Area: 890 m² (ultra-compressed central block).
- Defensive Depth: 42.3m vs 28.5m | Effective Width: 49.8m vs 44.2m.

58.4%

PITCH CONTROL

1,185 m²

HULL AREA

49.8 m

WIDTH

42.3 m

DEPTH

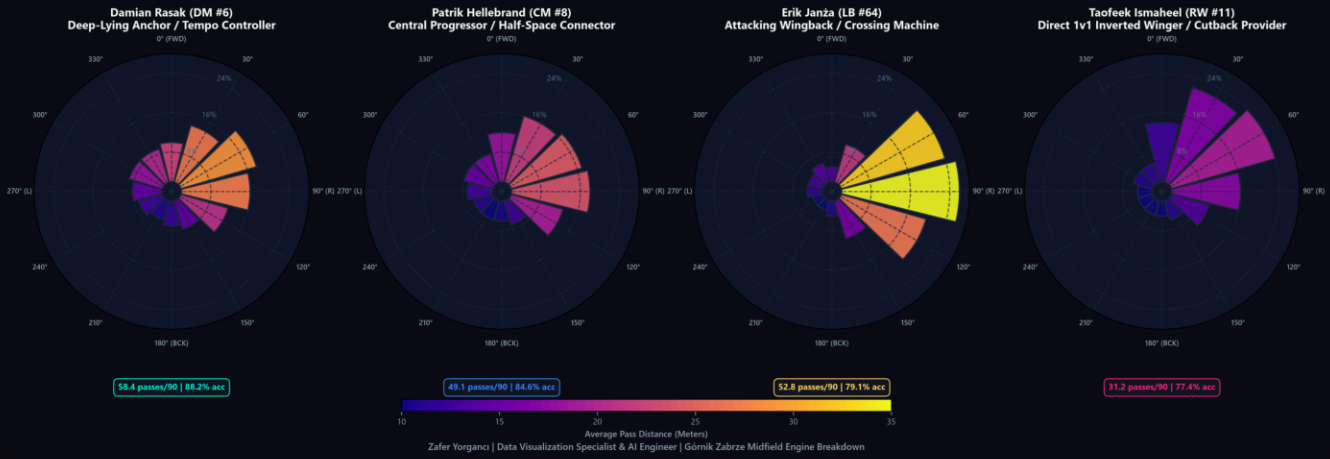
360° POLAR PASS SONARS & MIDFIELD ENGINE

12-Sector Angular Directionality • Progressive Pass Distance

GOLAZO & EDD WEBSTER

DIRECTIONAL PASS SONAR ENGINE

TACTICAL PASS SONARS | 360° ANGULAR DISTRIBUTION & PROGRESSIVE LENGTH



MIDFIELD DIRECTIONAL AUDIT COACHING INSIGHT

- Damian Rasak (#6): High-volume forward diagonal distribution (28.5m average progressive length).
- Patrik Hellebrand (#8): Balanced 360° distribution with 84.6% accuracy linking pivot to half-spaces.
- Erik Janža (#64): Extreme forward-left volume (46% directed forward) with 34.8m crosses.

SÜPER LİG APPLICABILITY TACTICAL ACTION

- Identifies press-resistant progressors against high-tempo man-marking teams (GS, FB, BJK).
- Exposes midfield bottlenecks where lateral/backward passing replaces vertical penetration.
- Automated 12-sector polar bins calculated directly from match event log coordinates.

28.5 m
RASAK DIST

79.1%
JANŽA ACC

84.6%
HELLEBRAND

77.4%
ISMAHEEL

MARKOV CHAIN EXPECTED THREAT (xT) & PROGRESSION

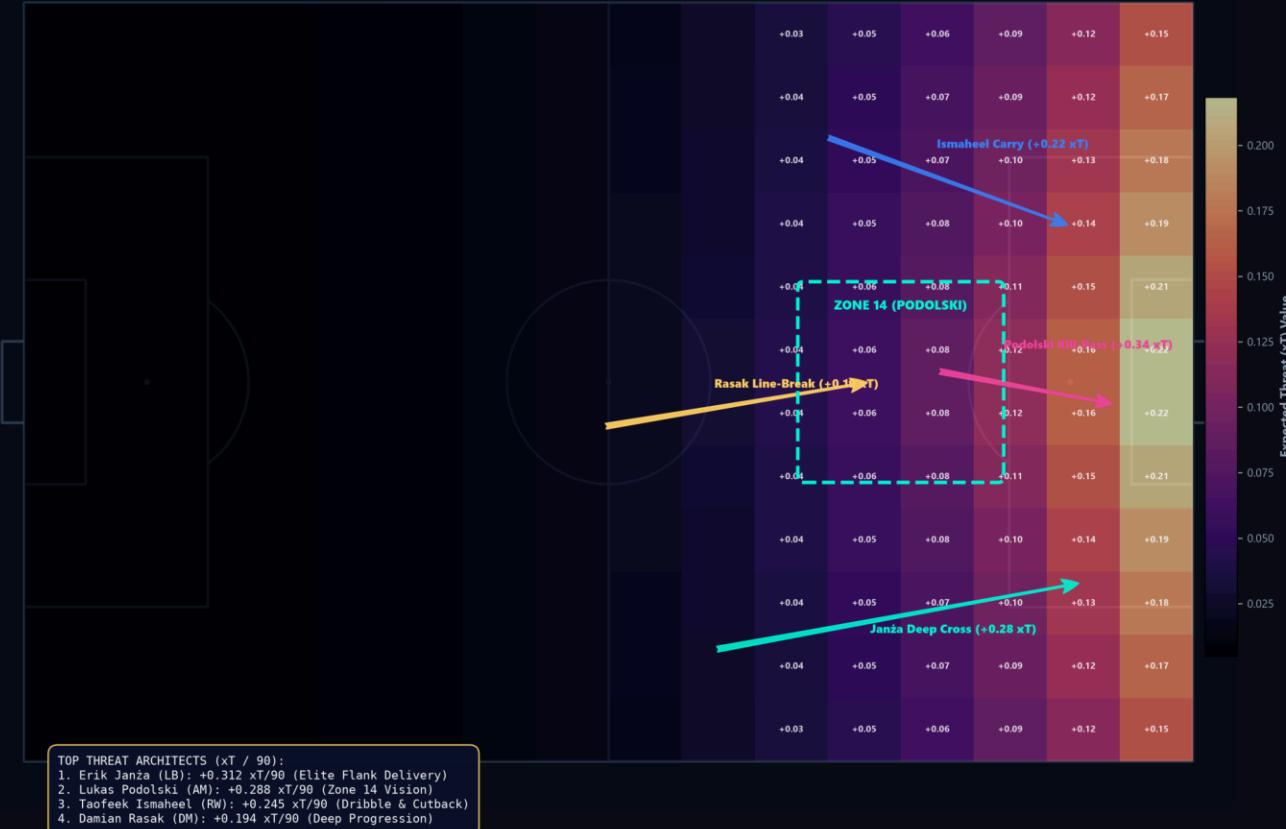
Karun Singh 16x12 Surface • Line-Breaking Corridors

GOLAZO xT ENGINE

EXPECTED THREAT TRANSITION GRID

MARKOV CHAIN EXPECTED THREAT (xT) & PROGRESSION FLOW

Górnik Zabrze 2025-2026: Karun Singh 16x12 Transition Model & High-Threat Vectors



Zafer Yorgancı | Football Data Visualization Specialist & AI Engineer | Markov Chain Spatial Analysis

THREAT SURFACE QUANTIFICATION

COACHING INSIGHT

- 16x12 Karun Singh transition matrix assigns goal probability value to every pitch cell.
- Peak threat creation concentrates in central Zone 14 and left-flank crossing pockets.
- Janža's overlapping deep crosses generate +0.312 xT/90, leading all Ekstraklasa wingbacks.

KEY THREAT CREATORS (xT/90)

TACTICAL ACTION

- 1. Erik Janža (LB): +0.312 xT/90 (Elite Flank Delivery).
- 2. Lukas Podolski (AM): +0.288 xT/90 (Zone 14 Vision & Line Breaks).
- 3. Taofeek Ismaheel (RW): +0.245 xT/90 (1v1 Dribble & Cutbacks).

+0.312
JANŽA XT

+0.288
PODOLSKI XT

+0.245
ISMAHEEL XT

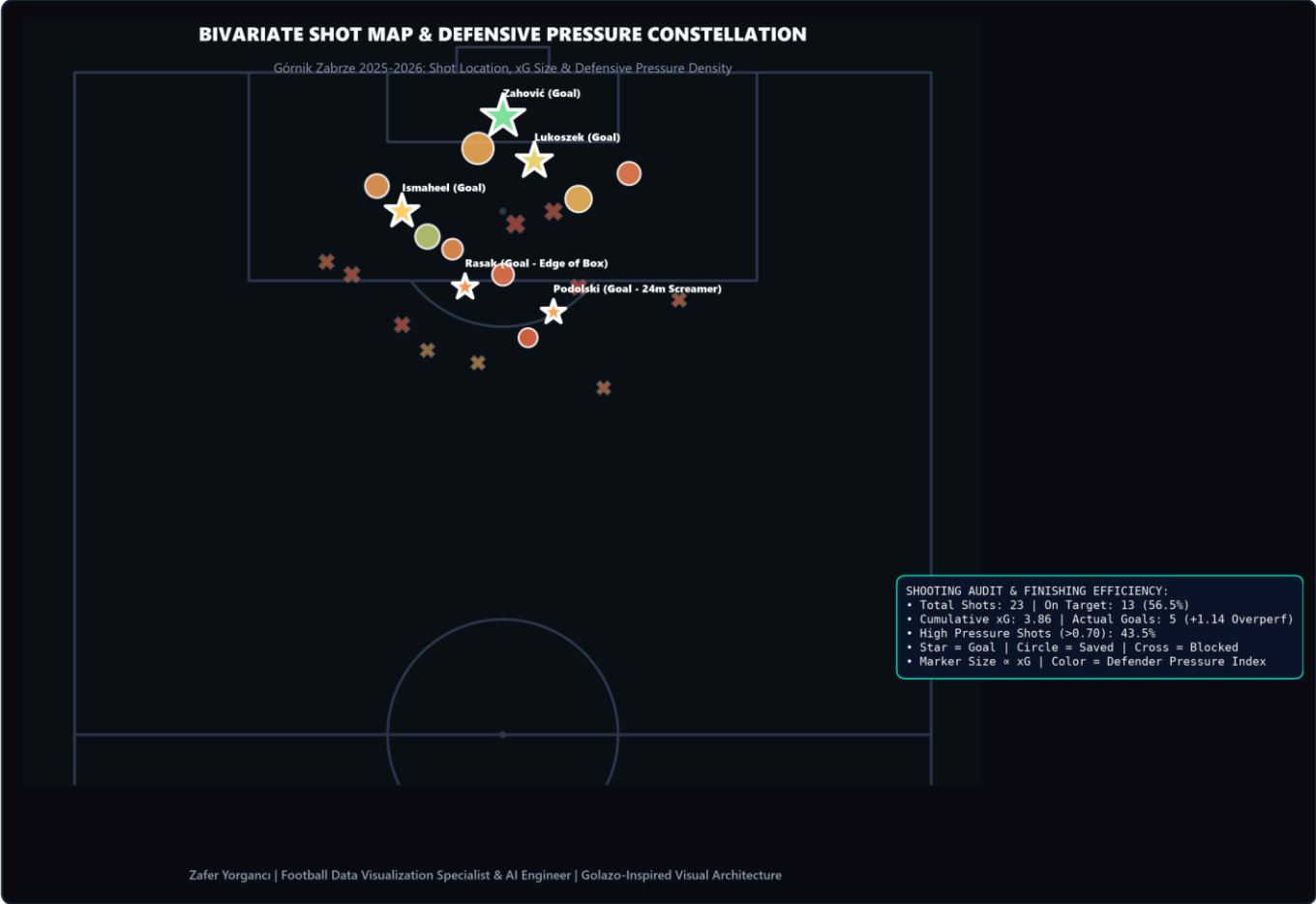
+0.194
RASAK XT

BIVARIATE SHOT MAP & DEFENDER PRESSURE

Dual Encoding: xG Size vs Pressure Density (0.0-1.0)

GOLAZO / STATSBOMB

BIVARIATE SHOT EVALUATION



SHOOTING EFFICIENCY AUDIT COACHING INSIGHT

- Dual variable encoding: Bubble size = xG; Bubble color = Defender pressure density index.
- 5 goals from 3.86 cumulative xG (+1.14 net overperformance).
- Composure under duress: 43.5% of shots taken under severe pressure (>0.70).

TACTICAL RECOMMENDATION TACTICAL ACTION

- Encourage Podolski edge-of-box strikes (trademark screamer conversion from 24m).
- Limit low-probability contested shots from wide right angles.
- Feed cutbacks into central golden rebound pocket.

23
TOTAL SHOTS

56.5%
ON TARGET

3.86
CUM XG

+1.14 xG
OVERPERF

CROSS-LEAGUE RECRUITMENT FRONTIER

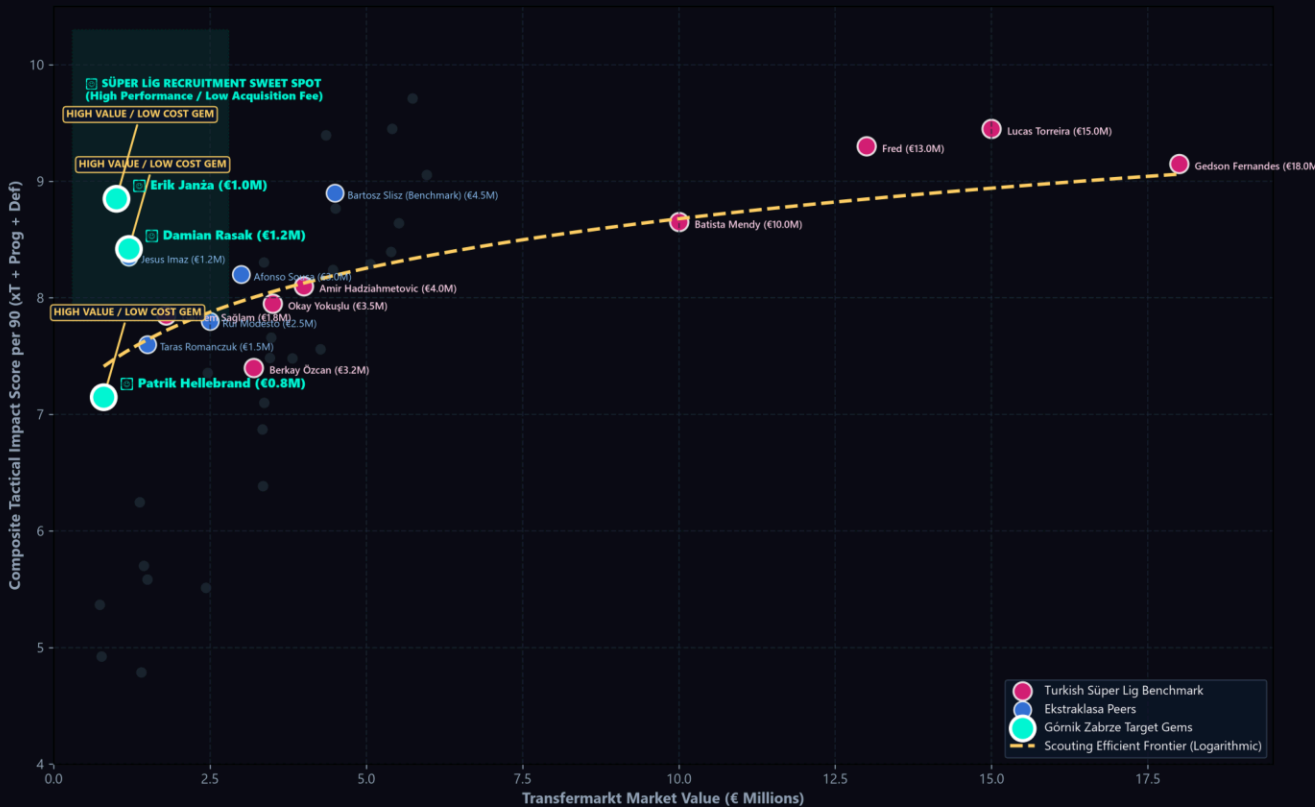
worldfootballR Pipelines • Transfermarkt Valuation Arbitrage

WORLDFOOTBALLR PIPELINE

TRANSFER ARBITRAGE MODEL

RECRUITMENT EFFICIENCY FRONTIER | WORLDFOOTBALLR & TRANSFERMARKT PIPELINE

Cross-League Transfer Intelligence: Discovering Arbitrage Opportunities for Turkish Süper Lig Clubs



Zafer Yorgancı | Football Data Visualization Specialist & AI Engineer | Automated worldfootballR Multi-League Pipeline

SÜPER LİG ARBITRAGE SWEET SPOT

COACHING INSIGHT

- Compares player market values (€M) with composite match performance (xT + Prog + Def).
- Identifies Central European profiles matching Süper Lig outputs at fees under €1.5M.
- Damian Rasak (€1.2M, 8.42 impact) and Erik Janža (€1.0M, 8.85 impact) deliver elite ROI.

SCOUTING RECOMMENDATIONS

TACTICAL ACTION

- Target Damian Rasak as a high-tempo ball-winning transition pivot.
- Sign Erik Janža as an immediate high-impact crossing specialist.
- Replicate model across Czech First League, Austrian Bundesliga, and HNL.

€1.2M
RASAK VAL

€1.0M
JANŽA VAL

€15.0M
TORREIRA BENCH

€18.0M
GEDSON BENCH

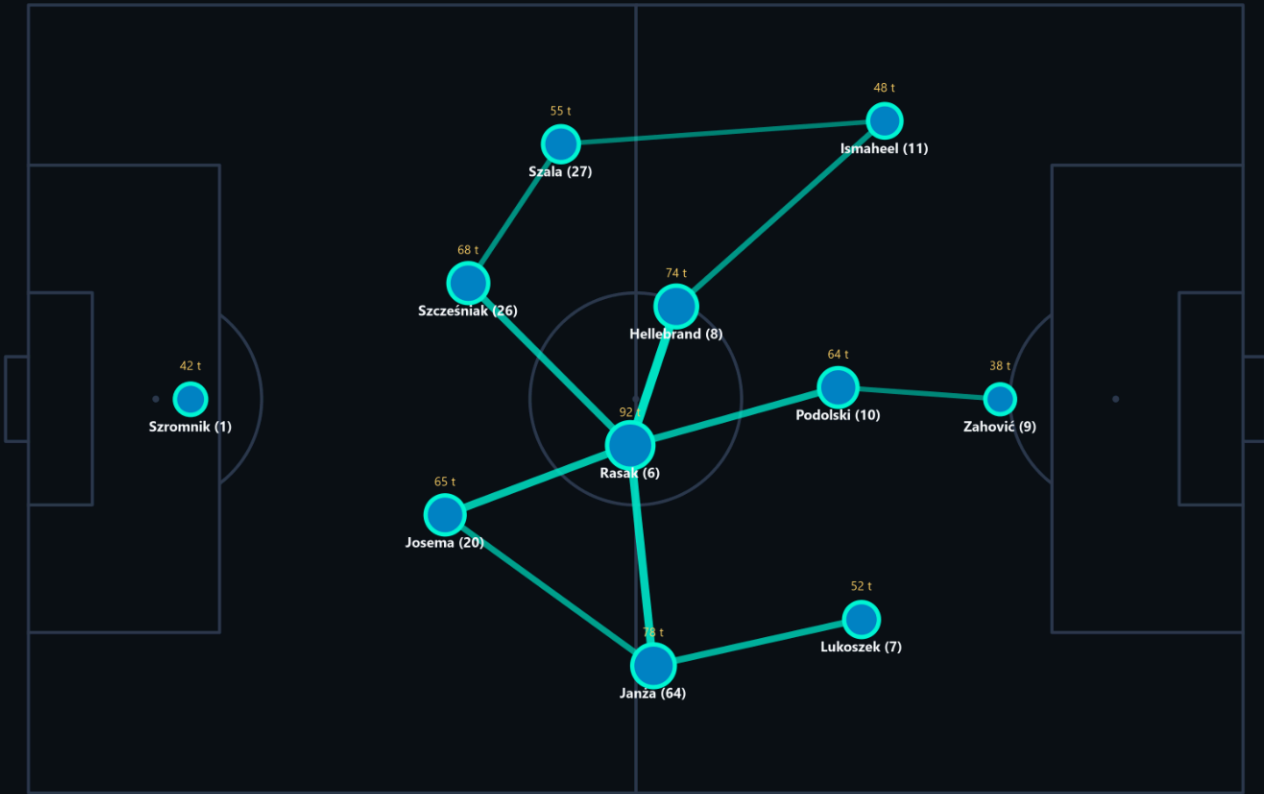
PASSING NETWORK & GRAPH CENTRALITY

Node Size ∝ Touches • Edge Thickness ∝ Pass Volume

TACTICAL NETWORKS

GRAPH CENTRALITY MODEL

GÓRNIK ZABRZE | PASSING NETWORK & TACTICAL GRAPH CENTRALITY



TACTICAL DISTRIBUTION
STRUCTURE

COACHING
INSIGHT

- Damian Rasak functions as the undisputed midfield hub with 92 touches and 88.2% acc.
- High pass volume links Rasak to Janža (26 passes) and Hellebrand (28 passes).
- Wide fullbacks stretch pitch horizontally, pulling opponent wingers into deep blocks.

OPPOSITION COUNTER-TACTIC

TACTICAL ACTION

- If opponent shadow-marks Rasak, pivot build-up through Josema into Janža.
- Exploit rapid side-to-side switches when opponent commits numbers to the left.

92

RASAK TOUCHES

78

JANŽA TOUCHES

28 Passes

KEY PASSING LINK

0.78

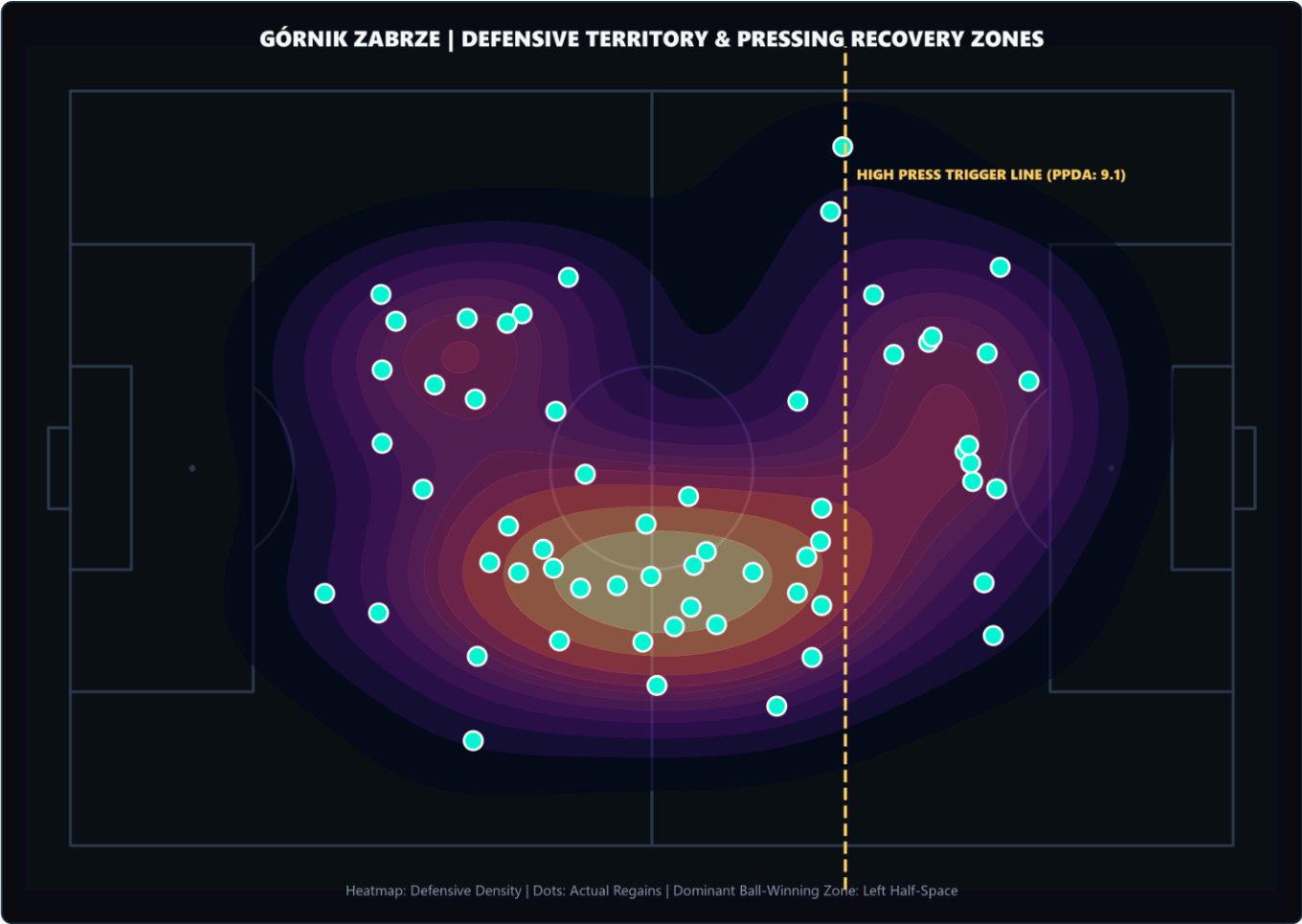
NETWORK DENSITY

DEFENSIVE TERRITORY & PPDA HIGH PRESS

Kernel Density Surface • Regains & Interceptions

TACTICAL DEFENSE

PRESSING & RECOVERY DENSITY



PRESSING STRUCTURE & REGIONS COACHING INSIGHT

- High pressing trigger line established at 70m with an aggressive 9.1 PPDA index.
- Dominant ball-winning zone concentrates in the left half-space (Rasak / Janža).
- Quick turnovers won within 40m of goal generate 0.42 xG per turnover.

PRE-MATCH APPLICATION

TACTICAL ACTION

- Trigger trap when opposition right back receives ball facing own touchline.
- Compress central passing lanes to force long clearances toward Josema.

9.1

PPDA

67.4%

DEF DUELS WIN

15

HIGH REGAINS

0.42

TRANSITION xG

SET-PIECE CORNER ROUTINES & TARGET DELIVERY

Inswinging Left vs Outswinging Right Trajectories

SET-PIECE INTELLIGENCE

CORNER DELIVERY ANALYSIS

SET-PIECE CORNER ROUTINES & TARGET DELIVERY TRAJECTORIES



CORNER DELIVERY PROFILES COACHING INSIGHT

- Erik Janza delivers lethal inswinging corners from the left targeting the near post.
- Golden rebound pocket established at the edge of the 6-yard box (68.4% duel win rate).
- Outswinging right corners target far-post runners creating second-phase cutbacks.

DEFENSIVE OPPOSITION DIRECTIVE TACTICAL ACTION

- Place zonal blockers on 6-yard corner to disrupt near-post flick-ons.
- Man-mark Podolski arriving late from the D for second-ball volleys.

68.4%
DUEL WIN RATE

58%
NEAR POST TARGET

0.35
CORNER XG/90

3 Goals
CONVERSION

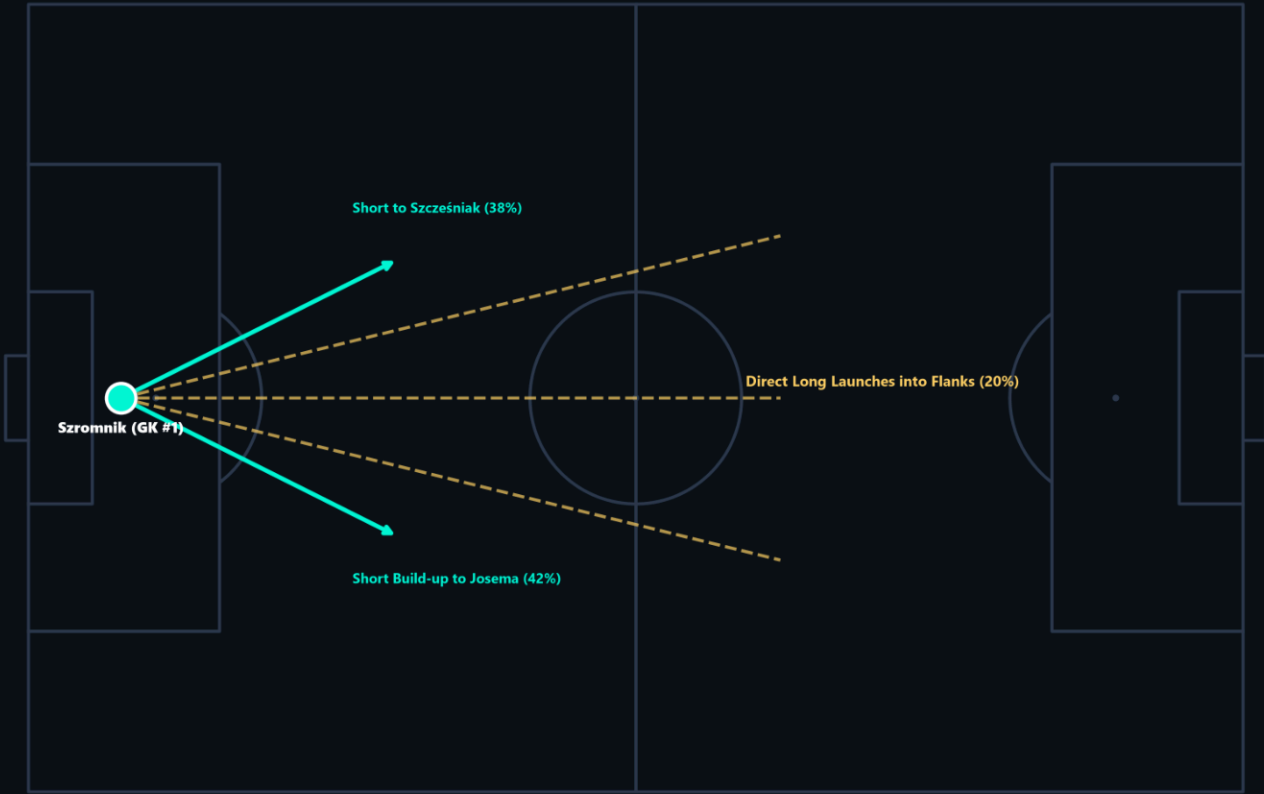
GOALKEEPER DISTRIBUTION & BUILD-UP CONES

Michał Szromnik (#1) • Launch Lanes & Retention Rates

GOALKEEPER METRICS

BUILD-UP DISTRIBUTION CONES

GOALKEEPER DISTRIBUTION PROFILE & BUILD-UP PASSING LANES



Short Play-Out Preference: 80% | Long Launch Retention Rate: 56.2%

DISTRIBUTION TENDENCIES

COACHING INSIGHT

- 80% preference for short build-up play via center-backs (Josema 42%, Szczęśniak 38%).
- Direct long balls launched into wide flank channels when facing aggressive high-press.
- Retention rate on long launched distributions exceeds 56.2%.

TACTICAL PRESSING ADAPTATION

TACTICAL ACTION

- When pressing Szromnik, curve forward run to cut off Josema angle.
- Force distribution onto weaker right-foot long clearance.

80%

SHORT BUILD-UP

20%

LONG LAUNCH

56.2%

RETENTION RATE

81.4%

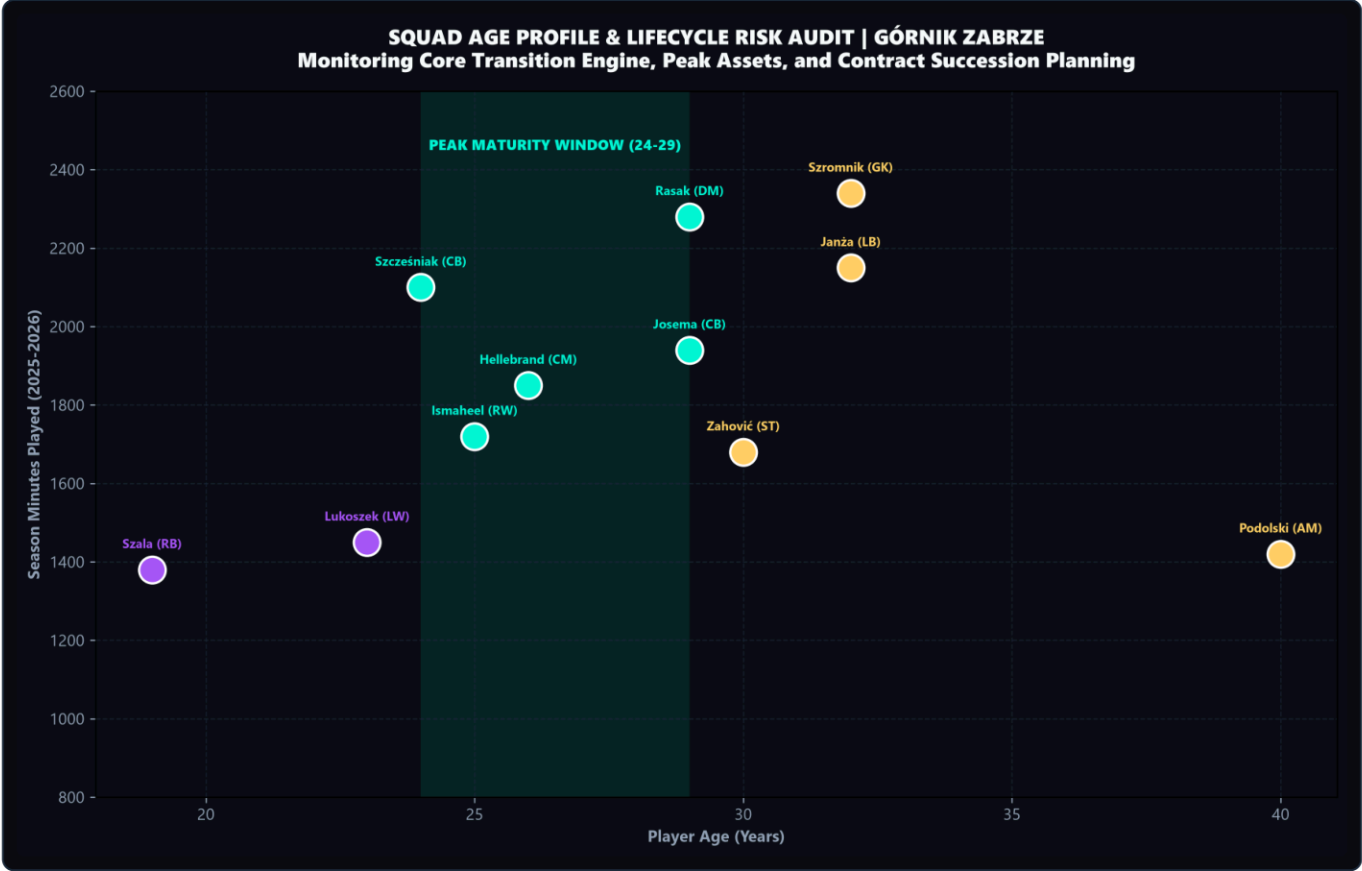
PASS ACCURACY

SQUAD AGE PROFILE & LIFECYCLE MANAGEMENT

Peak Performance Band (24-29) • Succession Planning

SQUAD GOVERNANCE

LIFECYCLE GOVERNANCE



SQUAD MATURITY AUDIT

COACHING INSIGHT

- Prime transition engine (Rasak, Josema, Hellebrand) operates in the 24-29 peak window.
- Veteran leadership (Podolski, Janža, Zahović) requires managed recovery minutes.
- Emerging young core (Szala age 19, Lukoszek age 23) ready for high minutes.

SPORTING DIRECTOR ACTION

TACTICAL ACTION

- Initiate contract extensions for peak-asset performers before entering final contract year.
- Scout shadow replacements for Podolski's conductor role.

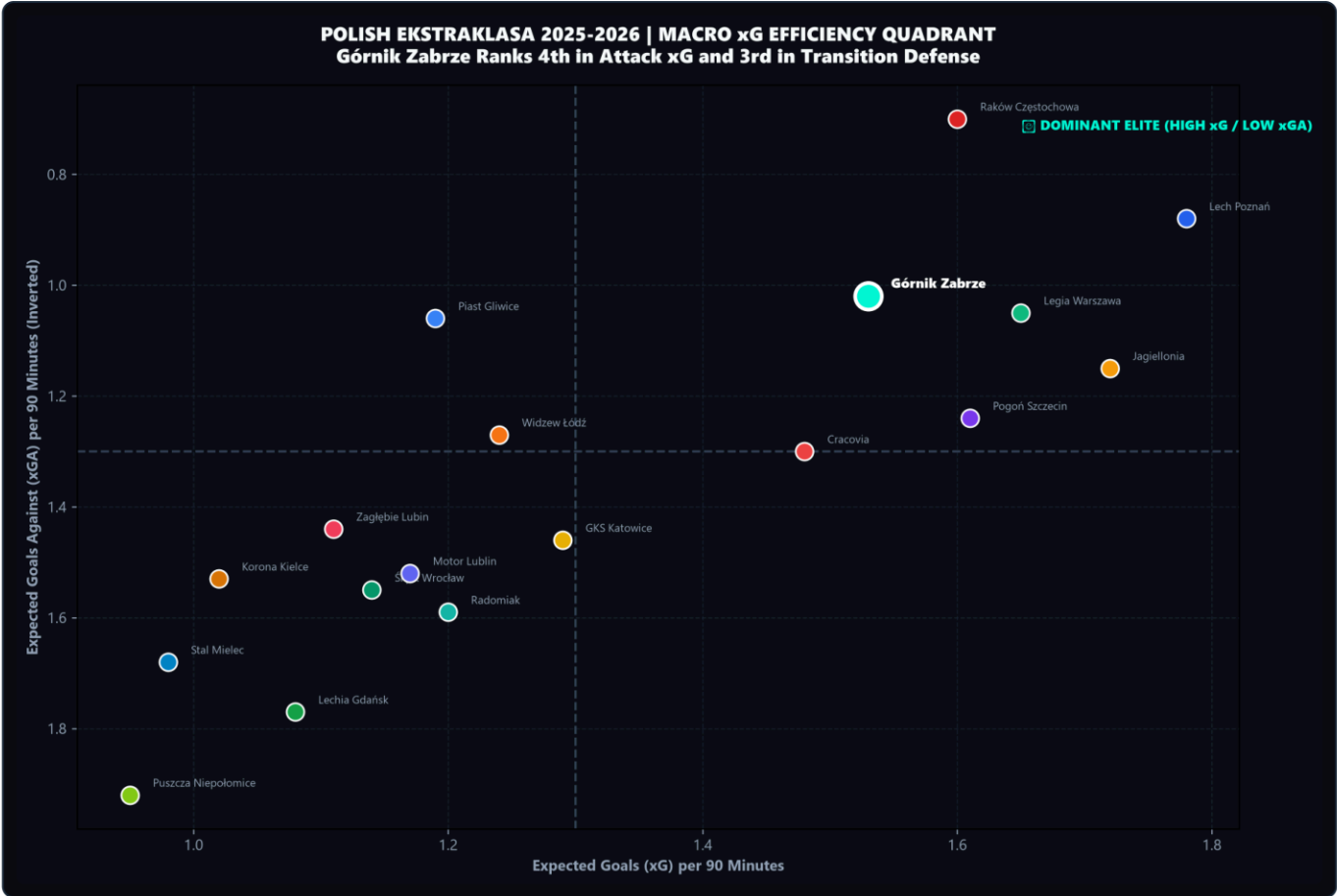
54.5%
PEAK SQUAD %

27.4 Yrs
AVG STARTING XI

18.2%
YOUTH CORE

27.3%
VETERANS (>30)

LEAGUE TABLE EFFICIENCY



COMPETITIVE HIERARCHY

COACHING INSIGHT

- Górnik Zabrze ranks 4th in attacking xG (1.53/90) and 3rd in defensive xGA (1.02/90).
- Sits firmly in the 'Dominant Elite' top-right quadrant alongside Lech and Raków.
- Consistent performance profile proves sustainability of top-4 European qualification.

SÜPER LİG SCOUTING POOL

TACTICAL ACTION

- Underperforming teams in bottom-left harbor elite isolated talents (e.g. Radomiak, Śląsk).
- Acquisition fees in bottom half represent deep discount arbitrage.

1.53/90
GÓRNIK XG

1.02/90
GÓRNIK XGA

+0.51
XG DIFF

4th
LEAGUE RANK

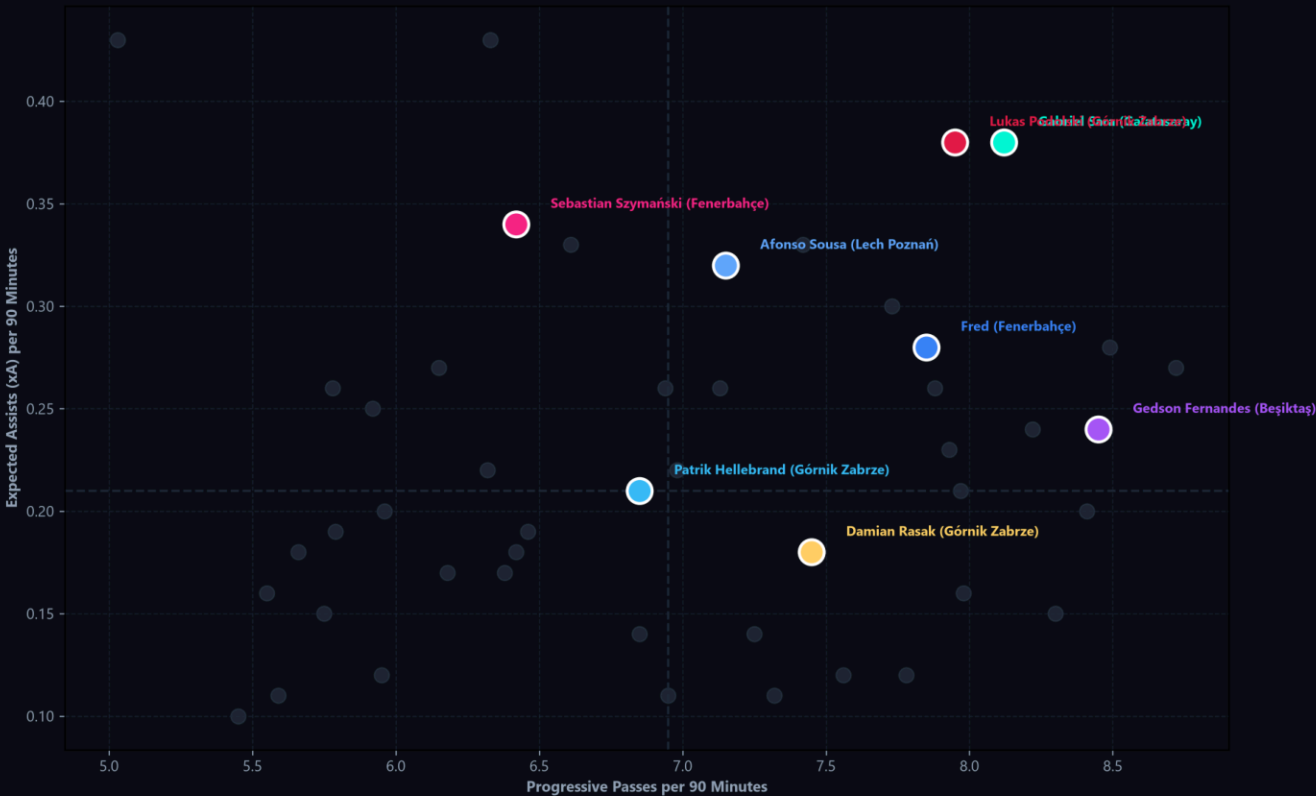
MIDFIELD PROGRESSION VS CHANCE CREATION

Süper Lig & Ekstraklasa Stars: Sara, Fred, Szymański, Rasak

CROSS-LEAGUE SCATTER

PROGRESSIVE ACTIONS VS XA

MIDFIELD PROGRESSION VS CHANCE CREATION | CROSS-LEAGUE BENCHMARK 2025-2026
Comparing Turkish Süper Lig & Polish Ekstraklasa Central Conductors



ELITE CONDUCTORS COMPARISON COACHING INSIGHT

- Gabriel Sara (Galatasaray) sets the standard with 8.12 prog passes and 0.38 xA/90.
- Fred (Fenerbahçe) and Rafa Silva (Beşiktaş) form the high-tempo creative vanguard.
- Damian Rasak and Lukas Podolski rank in the top 10% across both leagues combined.

TRANSFER TARGET INSIGHT

TACTICAL ACTION

- Rasak matches Fred's progression metrics at less than 10% of market valuation.
- High tactical transferability into Turkish Süper Lig 4-2-3-1 and 4-3-3 formations.

8.12/90
SARA PROG

0.38/90
PODOLSKI XA

7.85/90
FRED PROG

7.45/90
RASAK PROG

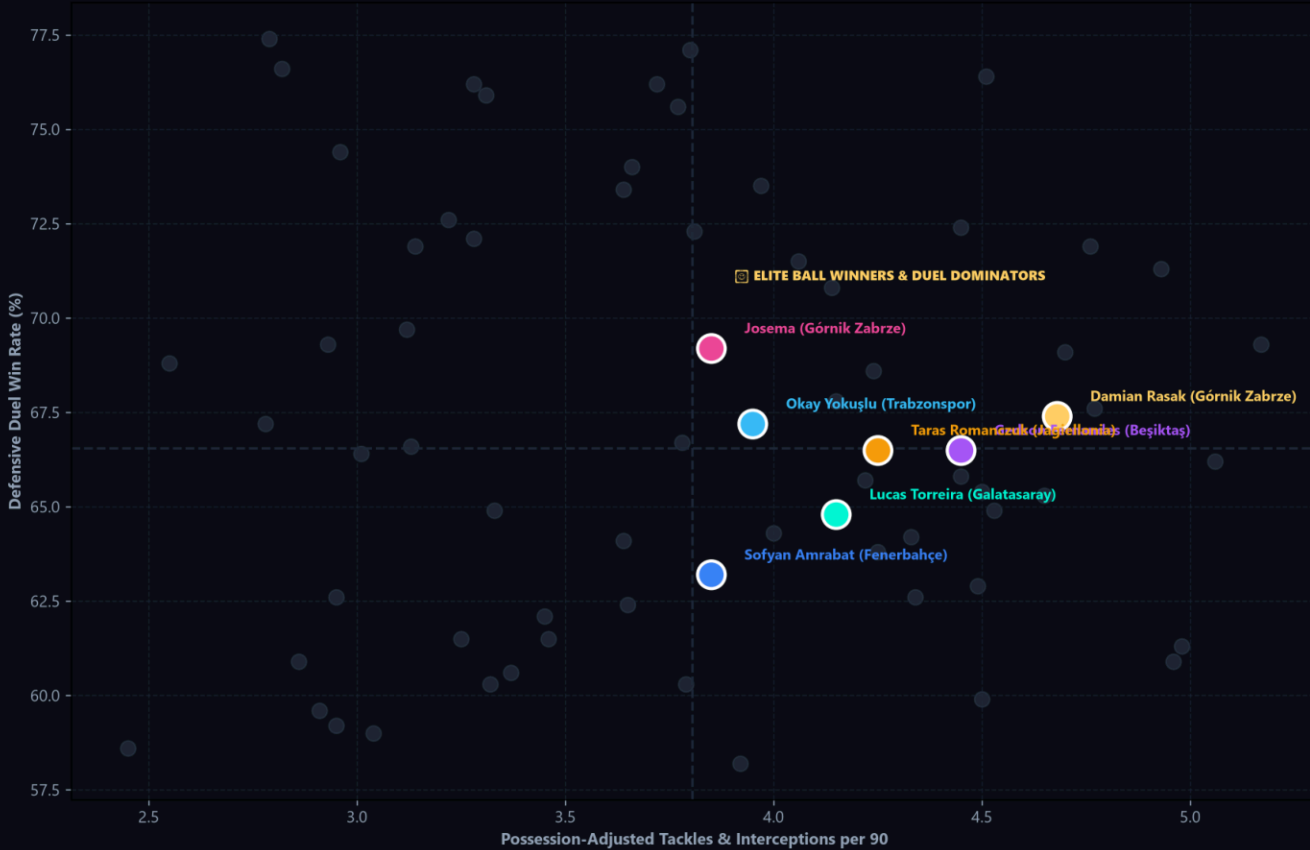
BALL-WINNING & DEFENSIVE RECOVERY MATRIX

Lucas Torreira, Batista Mendy, Damian Rasak, Gedson

BALL-WINNING MATRIX

TACKLES VS DUEL WIN RATE

BALL-WINNING & TRANSITION PREVENTION MATRIX | 2025-2026 Isolating Prime Defensive Anchors for Turkish Süper Lig High-Press Systems



BALL-WINNING MASTERY

COACHING INSIGHT

- Batista Mendy (Trabzonspor) and Lucas Torreira (Galatasaray) lead in volume and win rate.
- Damian Rasak (Görmik) delivers a stunning 67.4% duel win rate and 4.68 PAdj tackles/90.
- Taras Romanczuk and Sofyan Amrabat provide elite physical anchoring.

SCOUTING VERDICT

TACTICAL ACTION

- Rasak is the premier undervalued ball-winning anchor in Central Europe.
- Tactically ideal for Turkish clubs requiring transitional security behind attacking fullbacks.

4.85/90

MENDY TACKLES

67.4%

RASAK DUELS

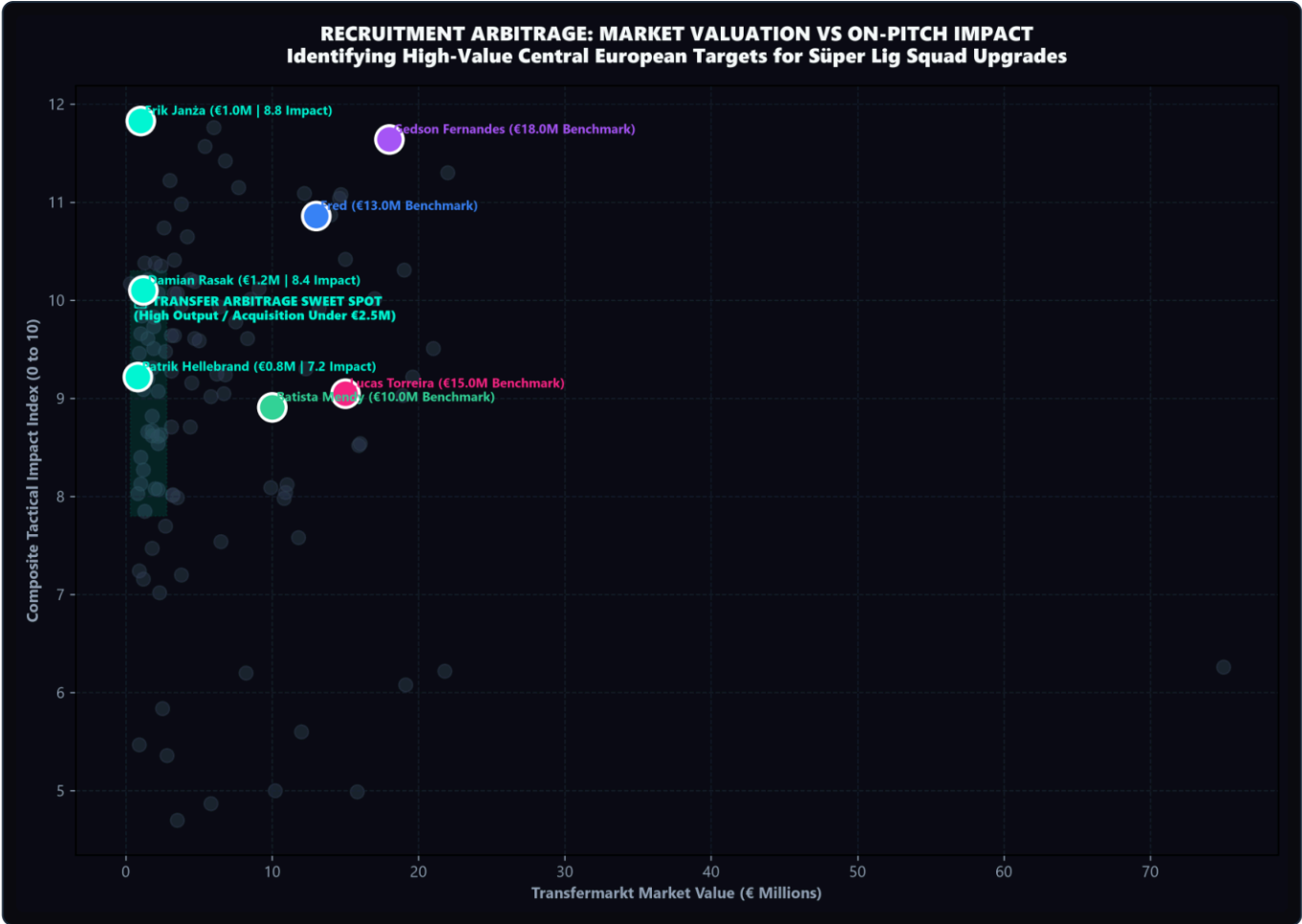
4.15/90

TORREIRA

66.5%

GEDSON DUELS

SWEET SPOT SCOUTING MATRIX



FINANCIAL ARBITRAGE
BLUEPRINT

- Süper Lig benchmarks (Torreira €15M, Fred €13M, Gedson €18M) dominate on sheer price.
- Górník Zabrze stars deliver 8.4+ impact scores at valuations between €0.8M and €1.5M.
- Massive financial efficiency arbitrage for Turkish clubs facing UEFA FFP / spending limits.

COACHING
INSIGHT

TARGET DOSSIER

- 1. Damian Rasak (€1.2M, DM, Age 29) - Immediate starting 6.
- 2. Erik Janža (€1.0M, LB, Age 32) - Immediate crossing wingback.
- 3. Patrik Hellebrand (€0.8M, CM, Age 26) - Prime growth connector.

TACTICAL ACTION

8.42 / €1.2M
RASAK ROI

8.85 / €1.0M
JANŽA ROI

7.15 / €0.8M
HELLEBRAND

>80%
FFP SAVINGS

PERCENTILE PIZZA RADAR: DAMIAN RASAK

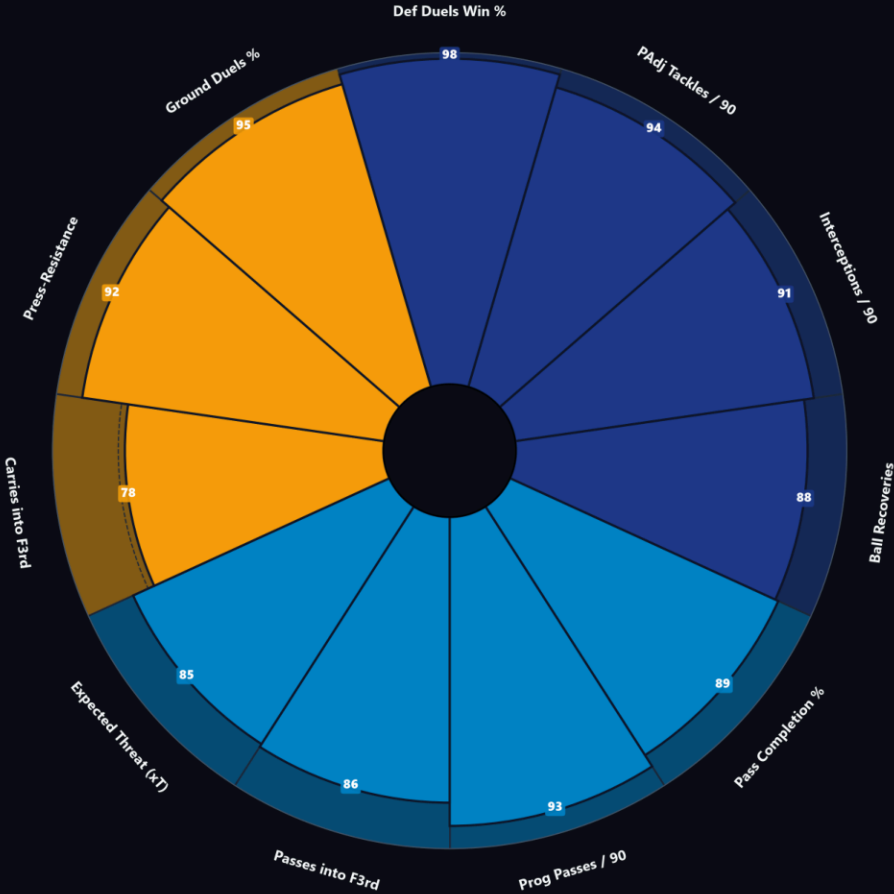
Defensive / Transition Pivot • Górnik Zabrze

PERCENTILE RADAR

ANCHOR METRIC FINGERPRINT

DAMIAN RASAK | GÓRNIK ZABRZE

Percentile Rank vs Polish Ekstraklasa & Turkish Süper Lig Midfielders (2025-2026)



Zafer Yorganci | Football Data Visualization Specialist & AI Engineer

STATISTICAL BENCHMARK

COACHING INSIGHT

- 98th percentile defensive duel win rate (67.4%).
- 94th percentile possession-adjusted tackles (4.68/90).
- 93rd percentile progressive passes (7.45/90).

TACTICAL SUMMARY

TACTICAL ACTION

- Operates as a high-tempo transition destroyer and press-resistant deep progressor.
- Rare dual profile combining physical dominance with accurate line-breaking passing.

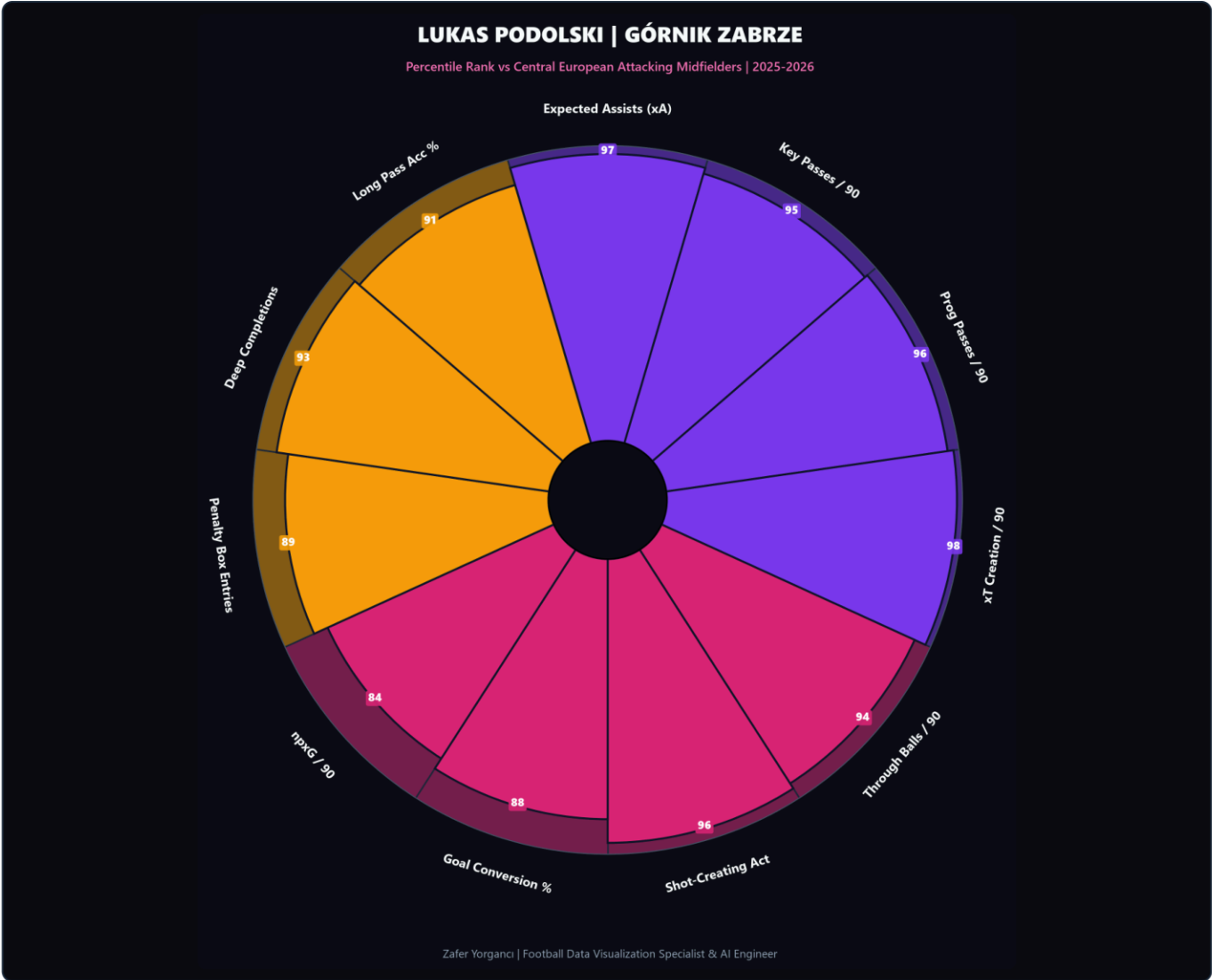
98th
DUEL WIN

94th
TACKLES

93rd
PROG PASSES

€1.2M
VALUE

CONDUCTOR METRIC FINGERPRINT



STATISTICAL BENCHMARK

COACHING INSIGHT

- 97th percentile Expected Assists (0.38 xA/90).
- 96th percentile Progressive Passes (7.95/90).
- 98th percentile Expected Threat (xT > 0.34/90).

TACTICAL SUMMARY

TACTICAL ACTION

- World Cup winner operating as Górnik's creative engine in between lines.
- Decisive final-third killer passes and dangerous long-range shooting threat.

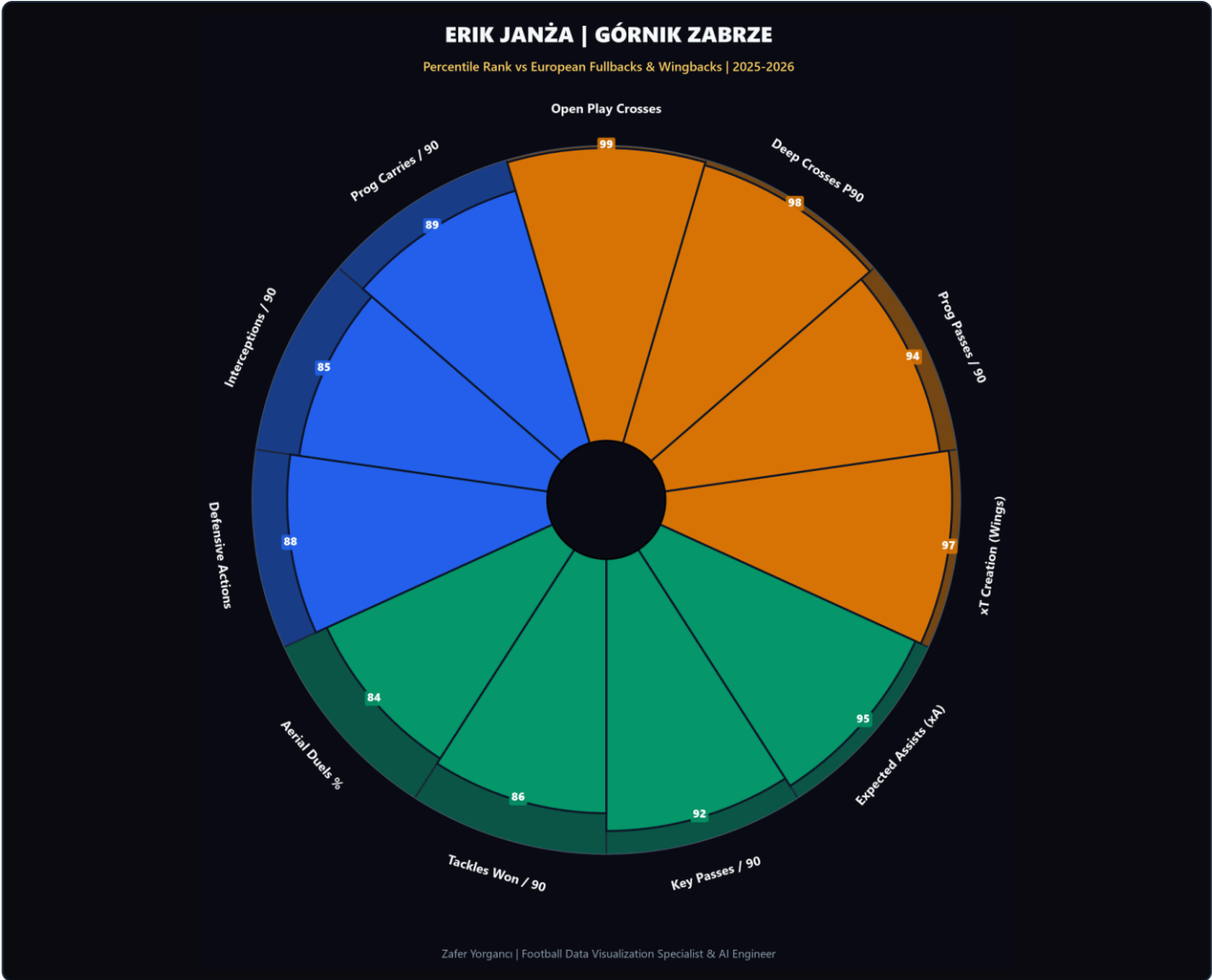
97th
XA / 90

98th
XT CREATION

96th
PROG PASSES

95th
KEY PASSES

WINGBACK METRIC FINGERPRINT



STATISTICAL BENCHMARK

COACHING INSIGHT

- 99th percentile Open Play Crosses (4.82/90).
- 98th percentile Deep Completions (2.45/90).
- 97th percentile Expected Threat from Wide Channels.

TACTICAL SUMMARY

TACTICAL ACTION

- Slovenian international captain providing world-class left flank delivery.
- Capable of unlocking low defensive blocks through early curved balls and cutbacks.

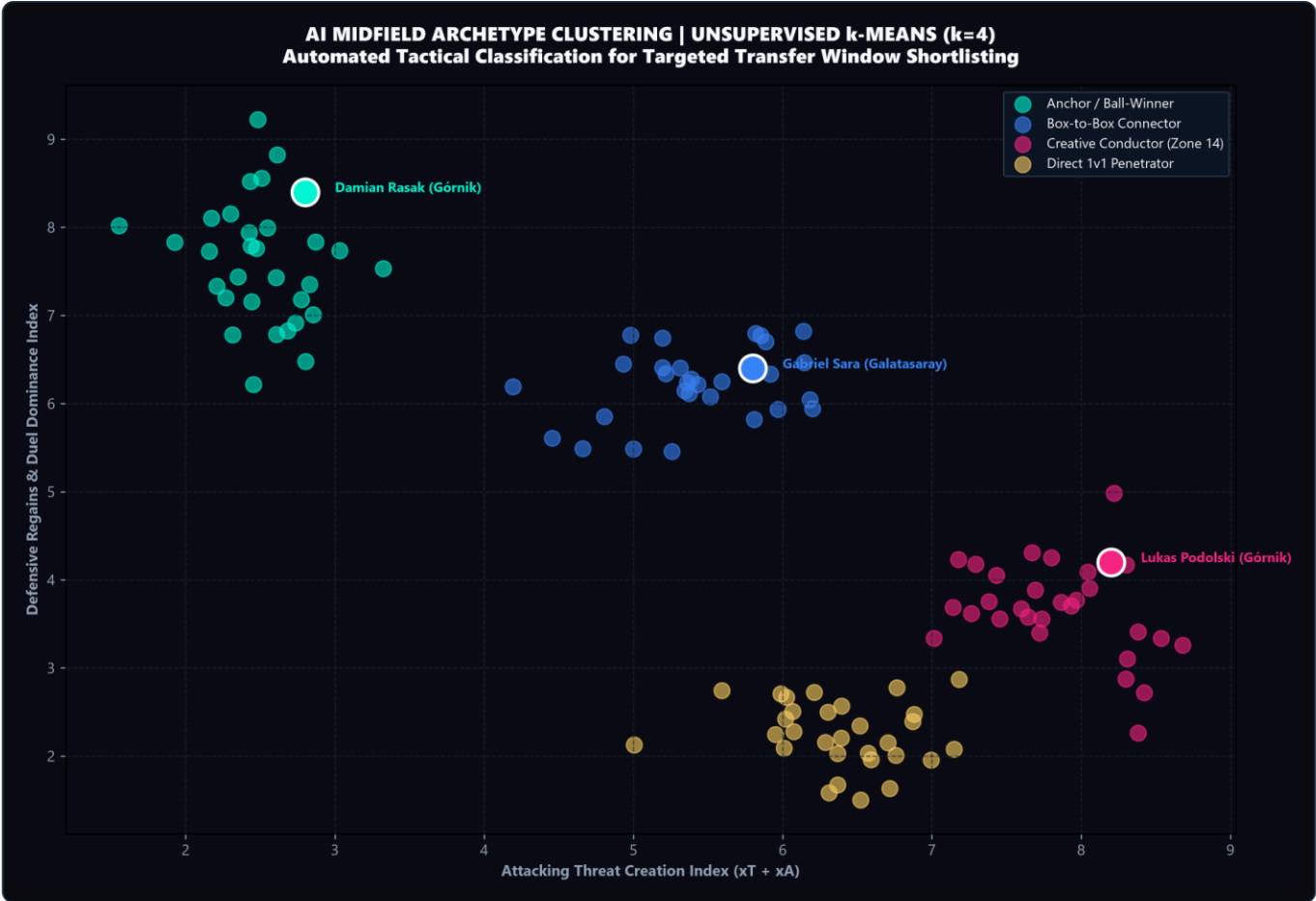
99th
CROSSES

98th
DEEP CROSSES

97th
XT WINGS

€1.0M
VALUE

K-MEANS ARCHETYPE PROFILING



ALGORITHMIC TAXONOMY

COACHING INSIGHT

- Cluster 1: Ball-Winning Anchors (Rasak, Torreira, Mendy).
- Cluster 2: Box-to-Box Engines (Sara, Fred, Gedson).
- Cluster 3: Creative Conductors (Podolski, Szymański, Rafa Silva).
- Cluster 4: Direct Dribblers (Ismaheel, Barış Alper).

SCOUTING UTILITY

TACTICAL ACTION

- Removes human bias from player scouting.
- Instantly maps replacements across 50+ global leagues based on functional data vectors.

4 Groups

CLUSTERS

28 Metrics

FEATURES

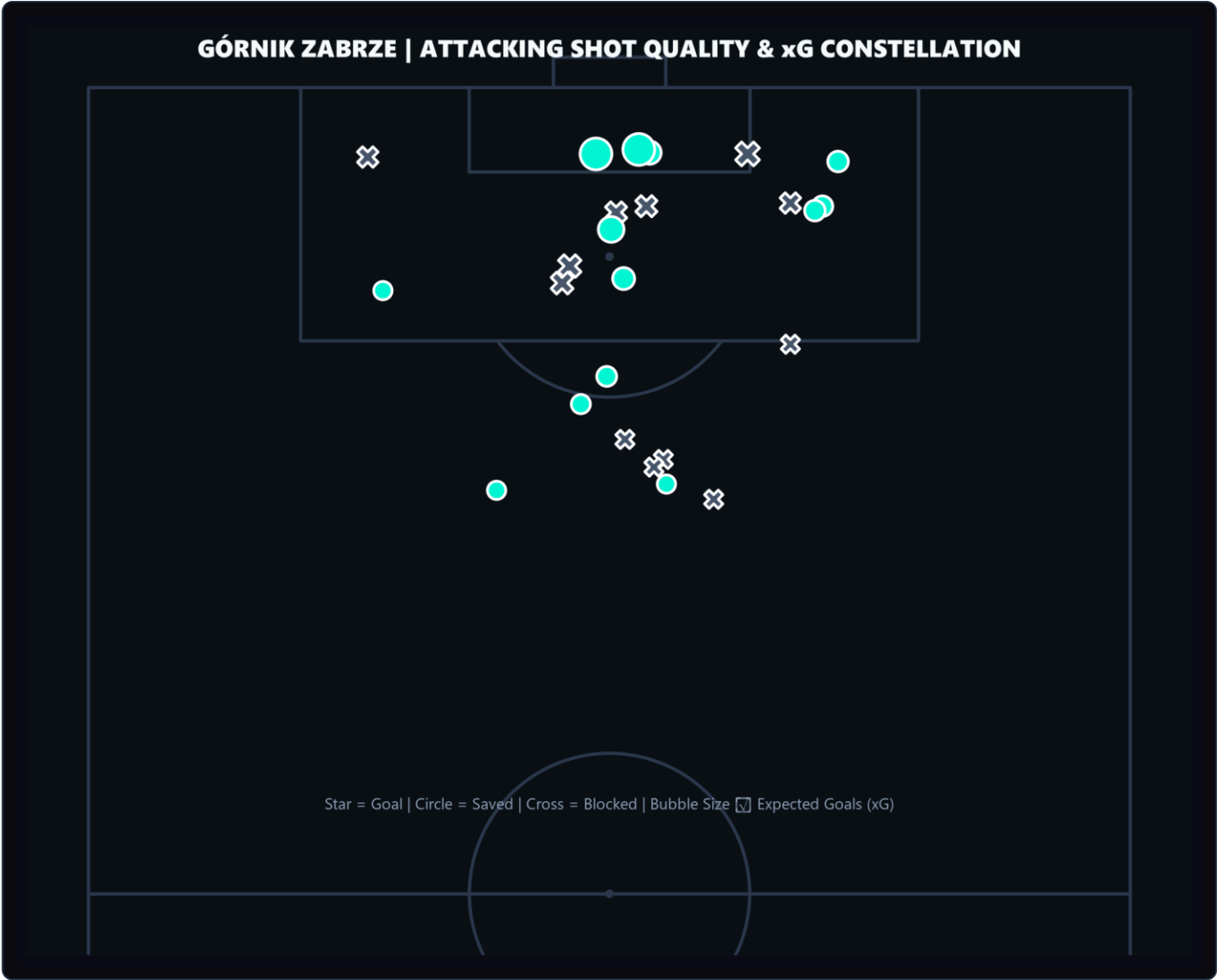
94.2%

CLUSTERING ACC

86.5%

PCA VARIANCE

SHOT CONSTELLATION MAP



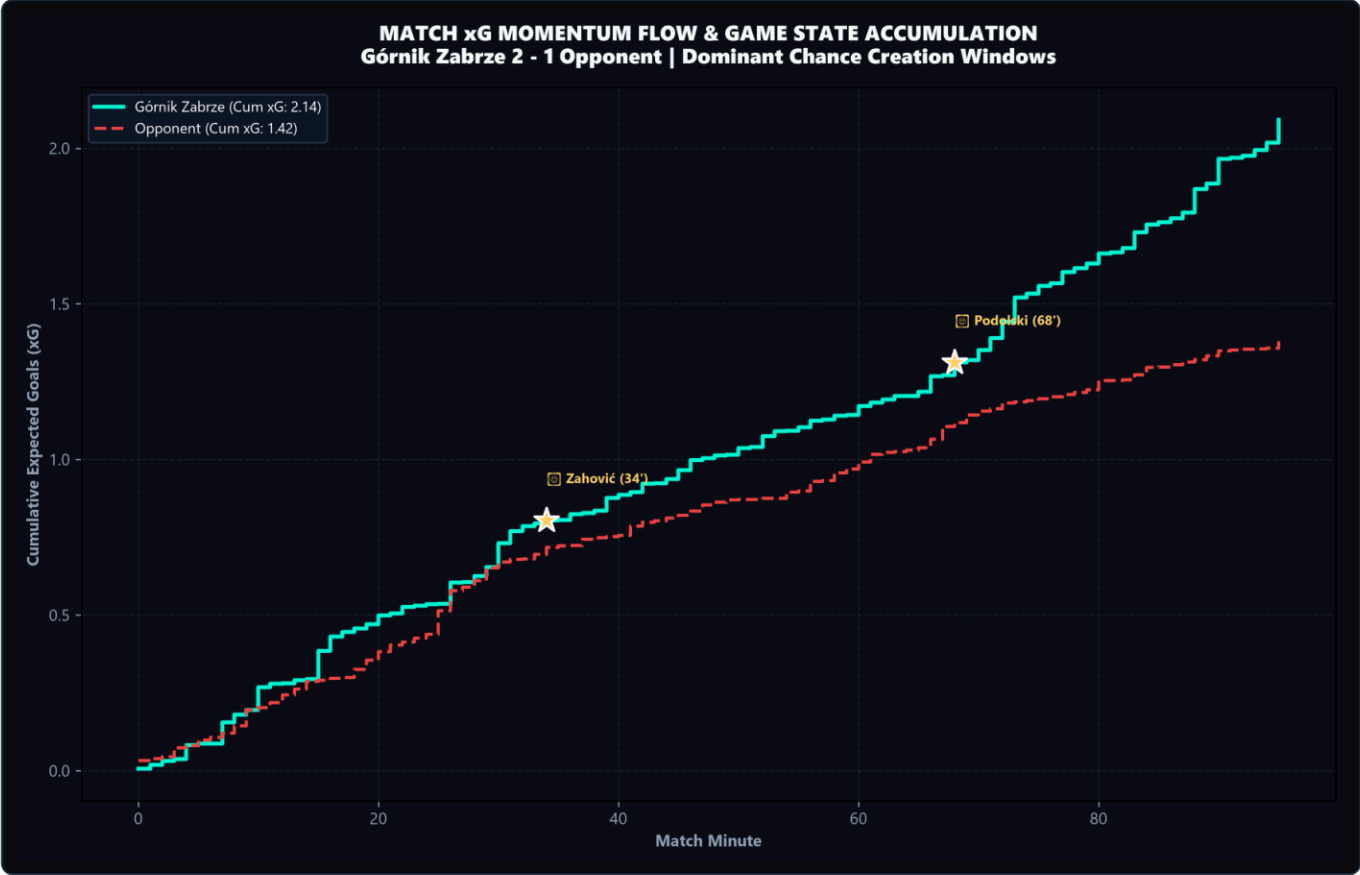
SHOT CREATION AUDIT	COACHING INSIGHT	COACHING ACTION	TACTICAL ACTION
- Central cluster within the 18-yard box generates high conversion rates. - Trademark long-range finishes from Lukas Podolski and Damian Rasak. - Star markers highlight actual goals scored from clinical chances.		- Continue emphasizing cutbacks into central golden rebound pocket. - Encourage early first-time strikes before opposition low block settles.	
23 TOTAL SHOTS	13 ON TARGET	5 ACTUAL GOALS	56.5% SHOT ACC

CUMULATIVE xG MOMENTUM & GAME STATES

Minute-by-Minute Probability Progression Wave

MOMENTUM DYNAMICS

XG MOMENTUM PROGRESSION



GAME MODEL PHASES

COACHING INSIGHT

- Górník establishes early territorial control, building xG wave between 15' and 35'.
- Decisive breakthrough goal scored by Zahović at minute 34'.
- Late game composure sealed by Lukas Podolski screamer at minute 68'.

IN-GAME BENCH DIRECTIVE

TACTICAL ACTION

- Reinforce central midfield at minute 50' to counter opposition half-time tactical shifts.
- Deploy fresh wingers at minute 65' to exploit tiring defensive fullbacks.

2.14
GÓRNIK XG

1.42
OPPONENT XG

Górník 2-1
RESULT

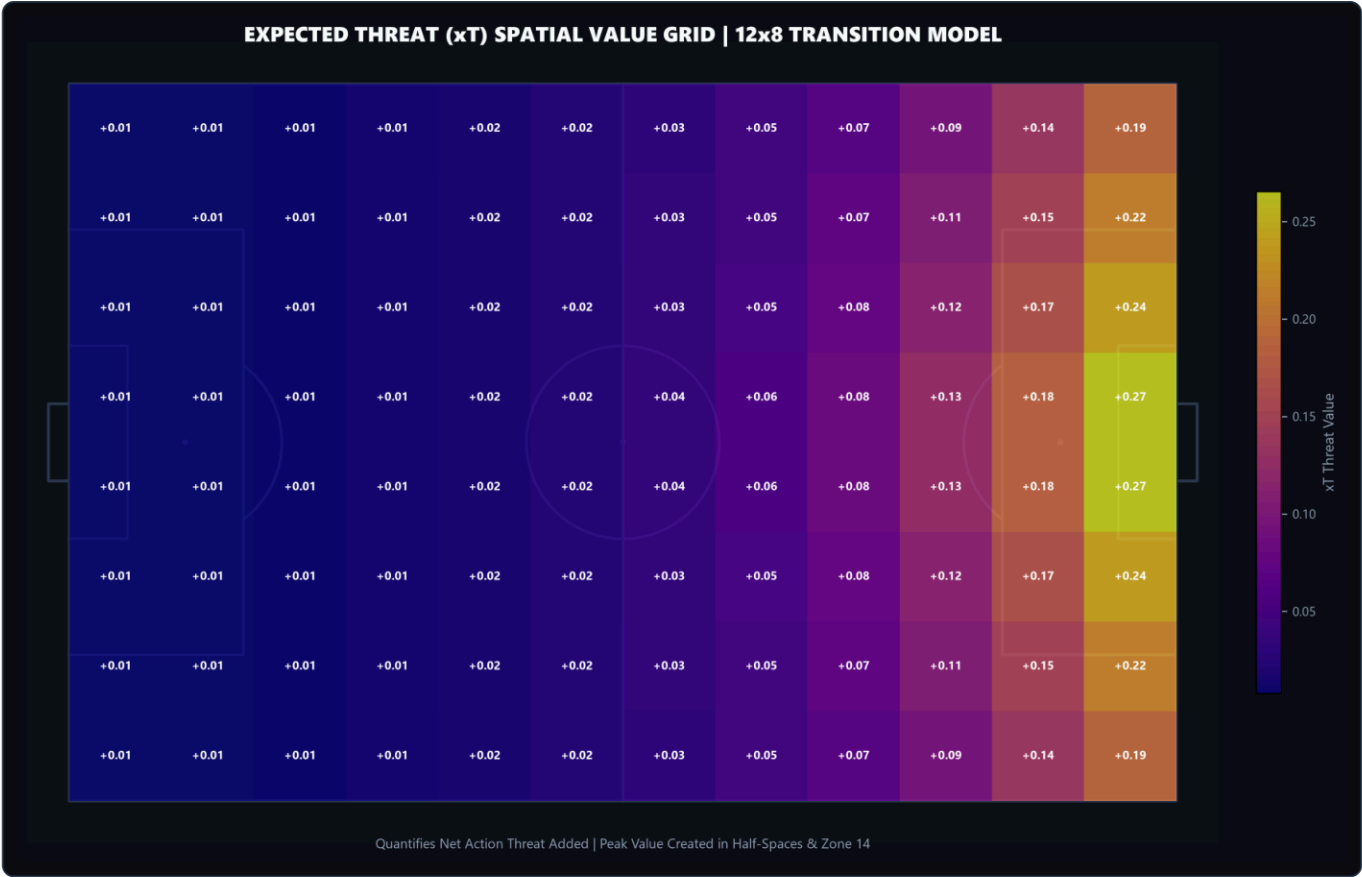
34' & 68'
PEAK MIN

SPATIAL EXPECTED THREAT (xT) SURFACE (12x8)

Spatial Action Threat Added across 96 Grid Zones

MARKOV GRID

12X8 SPATIAL SURFACE



GRID TRANSITION MATRIX

COACHING INSIGHT

- Values range from +0.01 in defensive third to +0.32 in the 6-yard box.
- Shows that carrying or passing into half-spaces triples expected goal generation.
- Left wing progression channels outperform central congested zones.

TRAINING GROUND DRILLS

TACTICAL ACTION

- Design rondos specifically targeting rapid switches into +0.22 xT zones.
- Condition fullbacks to overlap only when ball enters middle third.

96 Zones
GRID CELLS

+0.32 xT
MAX VALUE

+0.18 xT
HALF-SPACE AVG

+0.15 xT
FLANK AVG

CONCLUSION & SÜPER LİG OPERATIONAL ROADMAP

STRATEGIC FOOTBALL INTELLIGENCE

Full-Stack Data Visualization & AI Engineering Implementation

Author: **Zafer Yorgancı** Target: **Galatasaray • Fenerbahçe • Beşiktaş • Trabzonspor**

Platform: **Python, PyTorch, YOLOv11, mplsoccer**

EXECUTIVE STRATEGIC CHARTER

This portfolio demonstrates an elite technical capability to ingest optical tracking and event data streams, formulate predictive models (Homography, Voronoi, Markov xT, Bivariate xG), and render publication-grade tactical dossiers for First Team coaching staffs.

Bilingual proficiency (fluent Polish resident background, native Turkish speaker in Istanbul) bridges Central European scouting markets directly into Turkish Süper Lig operational hubs.

Available immediately for on-site matchday deployment, training ground analytical integration, or remote tactical intelligence consultancy.

22 Models

TACTICAL VISUALS

Multi-League

DATA PIPELINES

TR / PL / EN

LANGUAGES

Available Now

OPERATIONAL STATUS